



MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2015/2016
SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/COMPUTER
TECHNOLOGY

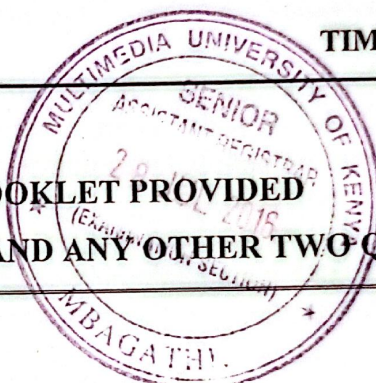
ICS 2104: OBJECT ORIENTED PROGRAMMING I

DATE: FRIDAY 22ND APRIL, 2016

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS



QUESTION ONE (THIRTY MARKS)

- [a]. Explain three benefits of object oriented software development methodology. [6 Marks]
- [b] Describe three primary features of object oriented programming. [6 marks]
- [c] Differentiate between the following:
 - i) Constructor and destructor [2 Marks]
 - ii) Class and object [2 Marks]
 - iii) Abstract class and base class [2 Marks]
- [d] During program design, explain why it is desirable to create loosely coupled and tightly cohesive methods. [4 Marks]
- [e] Explain three member access specifiers used in object oriented programming. [3 Marks]
- [f] Explain the importance of a *friend function*. [2 Marks]
- [g] Using appropriate example, explain operator overloading in object oriented programming. [3 Marks]

QUESTION TWO [TWENTY MARKS]

- [a] Distinguish between class declaration and class implementation. [4 Marks]
- [b] An object is an instance of a class. Explain this statement using appropriate example. [4 Marks]
- [c] Write a complete C++ program with class Rectangle and achieves the following:
 - i) Has a constructor with parameters width and height; and a destructor. [2 Marks]

S.m

- ii) Has a method to set new width and height for the Rectangle. [2 Marks]
- iii) Has an operation for computing area of a Rectangle. [2 Marks]
- iv) Instantiates class Rectangle with an object Rect1. [1 Marks]
- v) Creates a pointer to Rect1 called ptr. [2 Marks]
- vi) Use new operator to create another object. [1 Marks]
- vii) Have the two objects use their respective pointers to invoke get area method. [2 Marks]

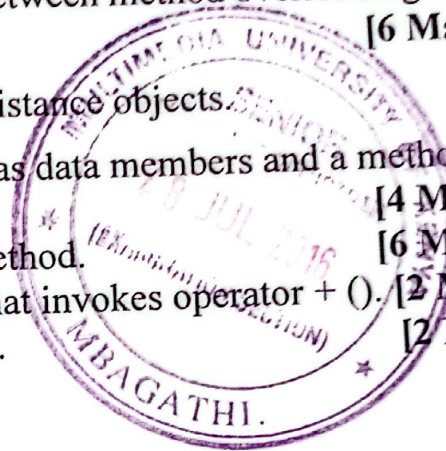
QUESTION THREE [TWENTY MARKS]

- [a] Explain three advantages of inheritance as used in object oriented programming. [3 Mark]
- [b] Choose any one of the three access specifiers, outline three implications of a derived class using this specifier for inheritance. [3 Marks]
- [c] A class Person has two fields: Id number and name; and two member functions, which and display the data fields. Write one complete C++ program that does the following:
 - i) Declare class Person and write its implementation. [4 Marks]
 - ii) Declare a derived class Customer that inherit the properties of Person; and has specific field balance due and two more member functions to set balance due and get balance due. [3 Marks]
 - iii) Write the implementation for class Customer. [2 Marks]
 - iv) Write a main () function that declares an array of one hundred Customer objects. Prompt the user for values for each Customer and displays the details of all the customers. [5 Marks]

QUESTION FOUR [TWENTY MARKS]

- [a] Differentiate between the following:
 - i) Virtual method and pure virtual method. [2 Marks]
 - ii) Early binding and late binding. [2 Marks]
- [b] Outline two overheads of using virtual methods. [2 Marks]
- [c] Imagine a publishing company that markets both book and audiocassette. Write one complete C++ program that implements the following:
 - i) Create a class called publication that stores the title (a string) and price of a publication and a getdata() function to get its data from the user at the keyboard, and a putdata() function to display the data. [2 Marks]
 - iii) From publication class derive two classes: book, which also has a page count; and tape, which also has playing time in minutes. [4 Marks]
 - iv) Write a main() program that creates an array of pointers to publication. [2 Marks]
 - v) In a loop, ask the user for data about a particular book or tape, and use new to create an object of type book or tape to hold the data. Put the pointer to the object in the array. [4 Marks]
 - vi) When the user has finished entering the data for all books and tapes, display the resulting data for all the books and tapes entered. [2 Marks]

QUESTION FIVE [TWENTY MARKS]

- [a] Using appropriate example, explain the difference between method overloading and method overriding. [6 Marks]
- [b] You are required to overload the '+' operator to add distance objects.
- i) Declare class distance with feet and inches as data members and a method operator+() that adds distance. [4 Marks]
 - ii) Write the implementation of operator+() method. [6 Marks]
 - iii) Create two objects of Distance in main () that invokes operator + (). [2 Marks]
 - iv) Explain the benefit of operator overloading. [2 Marks]
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UNIVERSITY EXAMINATIONS 2020/2021
SECOND YEAR FIRST SEMESTER SPECIAL / SUPPLEMENTARY EXAMINATION
FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER
SCIENCE/BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY

CCS 2211/BIT 2115: OBJECT ORIENTED PROGRAMMING I

DATE: 22ND JUNE 2022

TIME: 2 HOURS

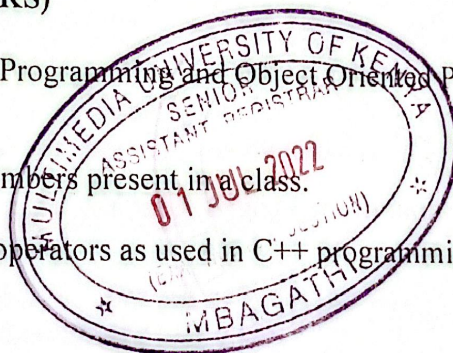
INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Differentiate between Structured Programming and Object Oriented Programming languages giving two example of each. (4 Marks)
- b) Differentiate between the two members present in a class. (2 Marks)
- c) Explain the use of the following operators as used in C++ programming. (3 Marks)
- i.) Dot(.)
 - ii.) Colon(:)
 - iii.) Double Colon (::)
- d) Explain the **three** access modifiers used in OOP. (6 Marks)
- e) Write a C++ program with class Circle that achieves the following requirements;
- i.) Has a default constructor that sets radius to zero, and has additional constructor with parameter for radius. (2 Marks)
 - ii.) Has a method to set new radius for the circle in which user is prompted to enter the radius (2 Marks)
 - iii.) Has an operation for computing area of circle making use of constant PI. (2 Marks)
 - iv.) Instantiate class with two objects, one to use default constructor and another to use parameterized constructor. (2 Marks)
 - v.) Display the initial area of each of the objects. (1 Mark)



- vi.) Set new radius for both objects.
- vii.) Displays the new area for both objects.

(1 Mark)

(1 Mark)

NB: this should be one program.

- a) Differentiate between the Global variables and Local variables.

(4 Marks)

QUESTION TWO (TWENTY MARKS)

- a) C++ data types are classified in three. State and give examples.

(3 Marks)

- b) Explain the following statements:

(2 Marks)

i. $7 == 5 ? 4 : 3$

ii. $5 > 3 ? a : b$

- c) Write a program that outputs "Hello, World!" n times (where n is a nonnegative integer that the user will input) with:

(9 Marks)

i. a for loop.

ii. while loop.

iii. a do...while loop.

- d) Rewrite the following program using the 'SWITCH' statement.

(6 Marks)

/* The traffic light program (nested ifs) */

```
include <iostream>
using namespace std;
```

```
int main ()
```

```
{
    char colour;
```

```
    /* ask user for colour */
```

```
    cout<<"Enter the colour of the light (R, G or Y): ";
    cin>>colour;
```

```
    /* test if colour is red */
```

```
    if (colour == 'r' || colour == 'R')
```

```
        cout<<"STOP!\n";
```

```
    else
```

```
        if (colour == 'y' || colour == 'Y') /* yellow colour test */
            cout<<"CAUTION!\n";
```

```
    else
```

```
        if (colour == 'g' || colour == 'G') /* green colour test */
            cout<<"GO!\n";
```

```
    else
```

```
        /* if not Y or G or R then invalid colour */
```

```
        cout<<"INVALID COLOUR!\n";
```


return (0);
}

QUESTION THREE (TWENTY MARKS)

- a) In a given bank, there are two types of bank accounts, namely Saving and Current. In both accounts, details of account number, holder name and balance must be kept. Both account have operations to check balance and to deposit. The withdraw operation is in both accounts but the conditions are different. In savings account, the withdrawal is only possible if balance is more than the amount being withdrawn. For current account, there is overdraft limit set for each client. Customers can withdraw beyond their balance but cannot go beyond the overdraft. Only Savings account earn interest on balance at a rate determined by the bank at the end of calendar year.
- i.) Draw class diagrams to represent the above scenario. (5 Marks)
 - ii.) Use C++ to implement the solution to the scenario. (15 Marks)

QUESTION FOUR (TWENTY MARKS)

- a) In a given hospital, patients are required to register upon arrival. The details taken include their name, gender and year of birth. Patients aged between 0 to 12 years should consult a pediatrician, otherwise they consult a physician. The patient is then issued with a prescription stating the drugs to be given and the amount to be paid. A system to be used in the hospital is required. The system should be able to give the details of the patient, the doctor to be consulted and prescription with cost.
- i.) Draw a class diagram to represent the above scenario (3 Marks)
 - ii.) Use C++ to implement the class (9 Marks)
- b) State the rules for naming variables and identifiers (4 Marks)
- c) Explain the meaning of *friendship* in classes and its importance in C++. (4 Marks)

QUESTION FIVE (TWENTY MARKS)

- a) Object Oriented Programming languages are preferred for large scale programming. Explain any four reasons that contribute to this. (4 Marks)
- b) Explain any *four* core concepts of Object Oriented Programming. (8 Marks)
- c) In every class, a constructor and destructor forms important component. Answer the following questions in relation to the two terminologies; (4 Marks)
- i.) Their characteristics
 - ii.) The roles they play
- d) State and explain the two ways you can define/declare constants. (4 Marks)



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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2019/2020
SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CCS 2213: OPERATING SYSTEMS II

DATE: 13TH DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- Explain two general roles of an operating system, and elaborate why these roles are important. [4 Marks]
- Explain the need for distributed operating systems and why they have become more important today [3 Marks]
- Explain processes in operating systems [2 Marks]
- Explain threads in operating systems [2 Marks]
- Explain is a race condition using an example. [3 Marks]
- Multi-programming (or multi-tasking) enables more than a single process to apparently execute simultaneously. How is this achieved on a single processor machine [2 Marks]
- State an advantages and disadvantage of user-level threads. [2 Marks]
- Explain a critical region and how it relates to controlling access to shared resources [2 Marks]
- What is *deadlock*? [2 Marks]
- Assuming the operating system detects the system is deadlocked, what can the operating system do to recover from deadlock? [2 Marks]
- Explain the importance of disabling interrupts during interprocess communication [3 Marks]
- Explain the message passing method in interprocess communication [3 Marks]

QUESTION TWO (TWENTY MARKS)

- a) Write some basic C code that creates a process. It should show whether it's a parent, child or no process created. [6 Marks]
- b) Explain four principal events that cause processes to be created [4 Marks]
- c) Explain four conditions for process termination [4 Marks]
- d) Explain two conditions under which a process may not be created [4 Marks]
- e) A *semaphore* is a blocking synchronization primitive. Explain how they work [2 Marks]

QUESTION THREE (TWENTY MARKS)

- a) Write some basic C code that creates a thread. [6 Marks]
- b) Using an application example of your choice explain how threads help in its task execution [4 Marks]
- c) Under what two conditions can a thread **not** be created [2 Marks]
- d) Explain three arguments of having threads [6 Marks]
- e) Explain two states that a thread can be in [2 Marks]

QUESTION FOUR (TWENTY MARKS)

- a) Write some basic C code to demonstrate RPC (remote procedure call) [8 Marks]
- b) Netflix is the world largest online movie provider. Explain how it uses replication to achieve a 24/7 uptime experience [5 Marks]
- c) Explain security management in distributed systems [4 Marks]
- d) Explain the concept of shared memory [3 Marks]

QUESTION FIVE (TWENTY MARKS)

- a) Explain three issues that developers in an organization must do to ensure a well structured InterProcess communication [6 Marks]
- b) Explain four conditions required in order to prevent race conditions. [2 Marks]
- c) Write some basic C code to demonstrate a semaphore [6 Marks]
- d) Explain the concept of distributed file systems [6 Marks]



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UNIVERSITY EXAMINATIONS (2019/2020)
SECOND YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN COMPUTER SCIENCE/SOFTWARE ENGINEERING

CCS 2211: OBJECT ORIENTED PROGRAMMING I

DATE: DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE [THIRTY MARKS]

- Compare and Contrast object oriented and structured programming techniques. (6 marks)
- Explain the following object oriented programming concepts, using appropriate examples: (6 marks)
 - Data abstraction
 - Inheritance
 - Polymorphism
- An object is an instance of a class. Explain this statement using appropriate example. (2marks)
- Explain the significance of scope resolution operator (::) (2 marks)
- Define a class student with the following specifications: (4 marks)

Adm_no integer

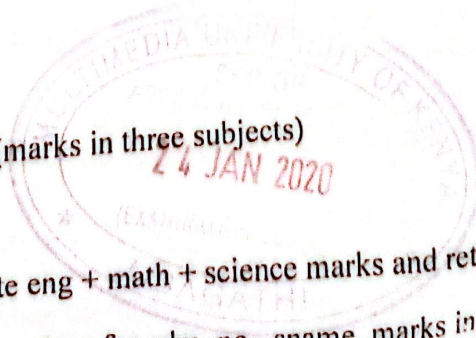
Sname 20 characters

Eng, math, science float (marks in three subjects)

Total float

ctotal() a function to calculate eng + math + science marks and return Total

takedata() function to accept values for adm_no, sname, marks in eng, math,



- showdata () function to display all the data members on the screen.
- f) Constructors and destructors are memory management methods in C++.
 - i) Distinguish between constructor and destructor. (2 marks)
 - ii) Using the class student above define: (3 marks)
 - A constructor having no parameters,
 - A Constructor which initialize data members to some values,
 - A destructor for the class.
 - g) Using appropriate examples, differentiate between function overloading and function overriding. (5 marks)

QUESTION TWO [TWENTY MARKS]

- a) Differentiate between the following:
 - i) Local variable and instance variable (2 marks)
 - ii) Abstract class and base class (2 Marks)
- b) Explain three class member access modifiers used in object oriented programming. (3 Marks)
- a) Write a complete C++ program with class Rectangle and achieves the following:
 - i) Has two constructors one without and one with parameters width and height; and a destructor. (3Marks)
 - ii) Has a method to set new width and height for the Rectangle. (2 Marks)
 - iii) Has an operation for computing area of a Rectangle. (2 Marks)
 - iv) Declares a pointer to Rectangle called ptr. (2Marks)
 - v) Uses new operator to create object Rect1. (2 Mark)
 - vi) Has Rect1 call the method in (iii) above using ptr. (2Marks)

QUESTION THREE [TWENTY MARKS]

- a) Cohesion and coupling are two design issues program designers need to deal with. Explain meaning of each, stating what should be promoted. (4 marks)
- b) Write a single C++ program that is able to achieve the following requirements:
 - i.) Creates a class called Student with data members: registration number, name, course and fee charged. (3 marks)
 - ii.) Use a method to initialize the data members for new students, where the user will be asked key in student details. (3 marks)
 - iii.) Use a method to enable student pay fees. The method should prompt the user to state amount of fees being paid, and then it should display the new fee balance after the payment. (3 marks)
 - iv.) Have a method for checking the current fee balance. The feedback should indicate whether student is in arrears, has cleared or has over paid. (3 marks)
 - v.) The main method should create a new student upon execution and then provide a list of operations to choose from. The list should have options to pay fee and check fee balance.

user should enter a number corresponding to one of these options failure to which they are informed their choice is invalid. (4 marks)

QUESTION FOUR [TWENTY MARKS]

- a) Explain two benefits of using Inheritance in Object Oriented programming. (4 marks)
- b) In a given company, there are two types of employees, casuals and permanent. For both, the company needs to capture their details such as name, ID number, gender and year of birth. Permanent employees have PF numbers but casuals don't. Both types of employees have salary that is taxed at 20% of the gross. For casuals, the gross pay depends on number of days worked for multiplied by daily rates that vary from time to time. The permanent staffs have basic salary plus allowances making up the gross. The allowances for permanent staff are not the same every month.
- i) Use class diagrams to model the scenario above. (4 marks)
- ii) Write a program in C++ that start by allowing the user to specify the type of employee to create, after which one can compute the employee's pay, showing the gross, tax payable and the net. (12 marks)

QUESTION FIVE [TWENTY MARKS]

- a) Explain dynamic binding as used in object oriented programming. (2 marks)
- a) Briefly describe the following, giving an example C++ program for each: - (9 Marks)
- i) Friend function
- ii) Virtual function
- iii) Function prototype
- b) Create an abstract class Shape with methods `calc_area` and `calc_perimeter`. Derive two classes, Circle with data radius and Rectangle with data length & breadth, from it. Calculate area and perimeter for the Circle and Rectangle. (9 Marks)



MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2016/2017

FIRST YEAR SECOND SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY &
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BIT 2109/ICS 2104 :

OBJECT ORIENTED PROGRAMMING 1

DATE: WEDNESDAY 28TH JUNE 2017



INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- Briefly explain why C++ is regarded as an object-oriented programming language
[2marks]
- Distinguish between overriding and overloading of functions in the context of object oriented programming
[2 marks]
- Write a sample C++ code that implements the method overloading concept. [4 marks]
- Using examples, differentiate between the following terms as used in C++ programming.
 - Class and Object
 - Encapsulation and Abstraction
 - Instance variables and Class variables[6 Marks]
- Write a C++ program which prompts the user to enter 2 integers, computes and prints their sum. In addition, the program should be able to determine the largest value between the integers.
[4 Marks]
- State two ways in which a constructor differs from the standard method
[2 Marks]
- Write a program to illustrate the use of constructor in C++ programming
[4 Marks]

(h) Define exception handling? [2 marks]

(i) Use an appropriate example program to illustrate how exceptions are handled in C++ programming [4 Marks]

QUESTION TWO (TWENTY MARKS)

(a) Write a C++ program using friend function to find the average of two numbers [4 marks]

(b) Using a nested loop, write a C++ program that produces the following output. [4 Marks]

```
*
* *
* * *
* * * *
* * * * *
```

(c) In a given company, there are two types of employees, casuals and permanent. For both, the company needs to capture their details such as name, ID number, gender and year of birth. Permanent employees have PF numbers but casuals don't. Both types of employees have salary that is taxed at 20% of the gross. For casuals, the gross pay depends on number of days worked for multiplied by daily rates that vary from time to time. The permanent staff have basic salary plus allowances making up the gross. The allowances for permanent staff are not the same every month.

(i) Use class diagrams to model the scenario above. [4 marks]

(ii) Write a program in C++ that allows a user to specify the type of employee to create, after which one can compute the employee's pay, showing the gross, tax payable and the net. [8 Marks]

QUESTION THREE (TWENTY MARKS)

(a) State the use of the 'this' keyword as used in object oriented programming [2 marks]

(b) Briefly describe the following, giving an example C++ program for each: - [9 marks]

- (i) Friend function
- (ii) Virtual function
- (iii) Function prototype

(c) Create an abstract class Shape with methods calc_area and calc_perimeter. Derive two classes, Circle with data radius and Rectangle with data length & breadth, from it. Calculate area and perimeter for the Circle and Rectangle. [9 Marks]

QUESTION FOUR (TWENTY MARKS)

- (a) Write a program in C++ that will achieve the following objectives:
- (i) Creates a class called Person with data members name and year of birth [2 Mark]
 - (ii) Overload the constructor, where one is parameterized and the other not. [4 Marks]
 - (iii) Create a method named getAge that prompts user to enter current year and return the age of the person. [2 Marks]
 - (iv) Create an object of type person that uses the parameterized constructor. [2 Marks]
- (b) Create a class Person that contains data members as Person Name, City, Mob_No. Write a C++ program to perform following functions:
- (i) To accept and display Person information
 - (ii) To search the mobile number of a given person
 - (iii) To search the Person details of a given mobile number

QUESTION FIVE (TWENTY MARKS)

- (a) Briefly state the difference between multiple inheritance and multi-level inheritance [4 Marks]
- (b) Distinguish between a virtual function and a pure virtual function as used in C++ programming. [4 Marks]
- (c) Write a C++ program demonstrating use of the pure virtual function [6 Marks]
- (d) Write a C++ program that can read name and gender of the user and write them in a text file called user.txt [6 marks]